*] kji.cpp

```c
#include <stdio.h>
int main()
{
    int marks[2][3][4];
    int i, j, k, total;
    float avg, sectionAvg[2], sum;
    int highest, count = 0;
    for(i = 0; i < 2; i++)
    {
        printf("\nSection %d\n", i + 1);

        for(j = 0; j < 3; j++)
        {
            printf("Enter marks of student %d in 4 subjects:\n", j + 1);
            total = 0;
            for(k = 0; k < 4; k++)
            {
                scanf("%d", &marks[i][j][k]);
                total = total + marks[i][j][k];
            }
            avg = total / 4.0;
            printf("Total = %d\n", total);
            printf("Average = %.2f\n", avg);

            if(avg >= 50)
                count++;
        }
    }
    for(i = 0; i < 2; i++)
    {
        highest = -1;
        sectionAvg[i] = 0;
        for(j = 0; j < 3; j++)
        {
            total = 0;
            for(k = 0; k < 4; k++)
                total = total + marks[i][j][k];

            sectionAvg[i] += total / 4.0;

            if(total > highest)
                highest = total;
        }
        sectionAvg[i] = sectionAvg[i] / 3;

        printf("\nSection %d highest total marks = %d\n", i + 1, highest);
        printf("Section %d average = %.2f\n",i + 1, sectionAvg[i]);
    }
    printf("\nStudents with average at least 50 = %d\n", count);
    if(sectionAvg[0] > sectionAvg[1])
        printf("Section 1 has the highest overall average.\n");
    else if(sectionAvg[1] > sectionAvg[0])
        printf("Section 2 has the highest overall average.\n");
    else
        printf("Both sections have the same average.\n");
    return 0;
}
```

C:\Users\HP\Documents\kji.e

```
Total = 244
Average = 61.00
Enter marks of student 3 in 4 subjects:
23 45 65 33
Total = 166
Average = 41.50

Section 2
Enter marks of student 1 in 4 subjects:
56 78 45 23
Total = 202
Average = 50.50
Enter marks of student 2 in 4 subjects:
67 55 88 96
Total = 306
Average = 76.50
Enter marks of student 3 in 4 subjects:
65 34 45 69
Total = 213
Average = 53.25

Section 1 highest total marks = 244
Section 1 average = 52.67

Section 2 highest total marks = 306
Section 2 average = 60.08

Students with average at least 50 = 5
Section 2 has the highest overall average.

--------------------------------
Process exited after 73.04 seconds with return value 0
Press any key to continue . . .
```

CamScanner

ji.cpp

```c
#include <stdio.h>
int main()
{
    int sales[2][3][4];
    int i, j, k;
    int total, branchTotal[2] = {0, 0};
    int weekTotal[4] = {0, 0, 0, 0};
    int productTotal, highest, bestProduct;

    for(i = 0; i < 2; i++)
    {
        printf("\nBranch %d\n", i + 1);
        for(j = 0; j < 3; j++)
        {
            printf("Product %d sales for 4 weeks:\n", j + 1);
            for(k = 0; k < 4; k++)
            {
                scanf("%d", &sales[i][j][k]);
                branchTotal[i] += sales[i][j][k];
                weekTotal[k] += sales[i][j][k];
            }
        }
    }
    for(i = 0; i < 2; i++)
    {
        highest = -1;
        bestProduct = 0;

        printf("\nBranch %d product totals:\n", i + 1);
        for(j = 0; j < 3; j++)
        {
            productTotal = 0;

            for(k = 0; k < 4; k++)
                productTotal += sales[i][j][k];

            printf("Product %d total = %d\n",j + 1, productTotal);

            if(productTotal > highest)
            {
                highest = productTotal;
                bestProduct = j + 1;
            }
        }

        printf("Highest-selling product = %d\n", bestProduct);
        printf("Branch total sales = %d\n", branchTotal[i]);
    }
    highest = weekTotal[0];
    bestProduct = 1;
```

CamScanner

```
45
46          printf("Highest-selling product = %d\n", bestProduct);
47          printf("Branch total sales = %d\n", branchTotal[i]);
48      }
49      highest = weekTotal[0];
50      bestProduct = 1;
51
52      for(i = 0; i < 4; i++)
53      {
54          printf("Week %d total sales = %d\n",i + 1, weekTotal[i]);
55
56          if(weekTotal[i] > highest)
57          {
58              highest = weekTotal[i];
59              bestProduct = i + 1;
60          }
61      }
62
63      printf("Week with highest sales = Week %d\n", bestProduct);
64
65      if(branchTotal[0] > branchTotal[1])
66          printf("Branch 1 sold the greatest quantity.\n");
67      else if(branchTotal[1] > branchTotal[0])
68          printf("Branch 2 sold the greatest quantity.\n");
69      else
70          printf("Both branches sold the same quantity.\n");
71
72      return 0;
73  }
```

CamScanner

```
Branch 1
Product 1 sales for 4 weeks:
200
400
980
780
Product 2 sales for 4 weeks:
340 760 98 678
Product 3 sales for 4 weeks:
450 670 890 300

Branch 2
Product 1 sales for 4 weeks:
450 300 400
78
Product 2 sales for 4 weeks:
300 200 100 120
Product 3 sales for 4 weeks:
400 800 600 56

Branch 1 product totals:
Product 1 total = 2360
Product 2 total = 1876
Product 3 total = 2310
Highest-selling product = 1
Branch total sales = 6546

Branch 2 product totals:
Product 1 total = 1228
Product 2 total = 720
Product 3 total = 1856
Highest-selling product = 3
Branch total sales = 3804
Week 1 total sales = 2140
Week 2 total sales = 3130
Week 3 total sales = 3068
Week 4 total sales = 2012
Week with highest sales = Week 2
Branch 1 sold the greatest quantity.

--------------------------------
Process exited after 61.89 seconds with return value 0
Press any key to continue . . .
```

CamScanner

```c
#include <stdio.h>

int main()
{
    float temp[2][3][4];
    float sum, avg, highest;
    int i, j, k, count = 0;

    for(i = 0; i < 2; i++)
    {
        printf("\nWard %d\n", i + 1);

        for(j = 0; j < 3; j++)
        {
            sum = 0;

            printf("Enter 4 temperature readings for patient %d:\n", j + 1);

            for(k = 0; k < 4; k++)
            {
                scanf("%f", &temp[i][j][k]);
                sum += temp[i][j][k];

                if(temp[i][j][k] > 37.5)
                    count++;
            }

            avg = sum / 4;

            printf("Patient %d average temperature = %.2f\n",j + 1, avg);

            if(avg > 38)
                printf("Patient %d average exceeds 38 C.\n", j + 1);
        }
    }

    for(i = 0; i < 2; i++)
    {
        highest = temp[i][0][0];

        for(j = 0; j < 3; j++)
        {
            for(k = 0; k < 4; k++)
            {
                if(temp[i][j][k] > highest)
                    highest = temp[i][j][k];
            }
        }

        printf("\nWard %d highest temperature = %.2f C\n",
               i + 1, highest);
    }

    printf("\nReadings above 37.5 C = %d\n", count);

    return 0;
}
```

```
C:\Users\HP\Documents\kji.e:

Ward 1
Enter 4 temperature readings for patient 1:
45 65 34 44
Patient 1 average temperature = 47.00
Patient 1 average exceeds 38 C.
Enter 4 temperature readings for patient 2:
33 65 34 22
Patient 2 average temperature = 38.50
Patient 2 average exceeds 38 C.
Enter 4 temperature readings for patient 3:
54 55 34 55
Patient 3 average temperature = 49.50
Patient 3 average exceeds 38 C.

Ward 2
Enter 4 temperature readings for patient 1:
45 67 55 43
Patient 1 average temperature = 52.50
Patient 1 average exceeds 38 C.
Enter 4 temperature readings for patient 2:
33 22 34 56
Patient 2 average temperature = 36.25
Enter 4 temperature readings for patient 3:
67 55 45 33
Patient 3 average temperature = 50.00
Patient 3 average exceeds 38 C.

Ward 1 highest temperature = 65.00 C

Ward 2 highest temperature = 67.00 C

Readings above 37.5 C = 15

--------------------------------
```

CamScanner

kji.cpp

```c
#include <stdio.h>
float calculateBill(int units, int category)
{
    float bill;
    if(category == 1)
    {
        if(units <= 100)
            bill = units * 15;
        else
            bill = 100 * 15 + (units - 100) * 20;
    }
    else
    {
        bill = units * 25;
    }
    return bill;
}

int main()
{
    int units, category;
    float bill, surcharge = 0, finalBill;

    printf("Enter units consumed: ");
    scanf("%d", &units);

    printf("Select customer category:\n");
    printf("1. Domestic\n");
    printf("2. Commercial\n");
    printf("Enter your choice: ");
    scanf("%d", &category);

    if(units < 0 || (category != 1 && category != 2))
    {
        printf("Invalid input!\n");
        return 0;
    }
    bill = calculateBill(units, category);

    if(bill > 5000)
        surcharge = bill * 0.05;

    finalBill = bill + surcharge;

    printf("\nUnits Consumed = %d\n", units);
    printf("Bill Amount = Rs. %.2f\n", bill);
    printf("Surcharge = Rs. %.2f\n", surcharge);
    printf("Final Payable Amount = Rs. %.2f\n", finalBill);

    return 0;
}
```

C:\Users\HP\Documents\kji.e

```
Enter units consumed: 400
Select customer category:
1. Domestic
2. Commercial
Enter your choice: 1

Units Consumed = 400
Bill Amount = Rs. 7500.00
Surcharge = Rs. 375.00
Final Payable Amount = Rs. 7875.00

--------------------------------
Process exited after 33.3 seconds with return value 0
Press any key to continue . . .
```

CamScanner

```c
#include <stdio.h>
float calculateBonus(float salary, int years)
{
    float bonus;

    if(years > 5)
        bonus = salary * 0.15;
    else if(years >= 3)
        bonus = salary * 0.10;
    else
        bonus = salary * 0.05;

    return bonus;
}

int main()
{
    float salary, bonus, total, tax = 0, finalSalary;
    int years;

    printf("Enter basic salary: Rs. ");
    scanf("%f", &salary);

    printf("Enter years of service: ");
    scanf("%d", &years);

    if(salary < 0 || years < 0)
    {
        printf("Invalid input!\n");
        return 0;
    }

    bonus = calculateBonus(salary, years);

    total = salary + bonus;

    if(total > 100000)
        tax = total * 0.05;

    finalSalary = total - tax;

    printf("\nBasic Salary = Rs. %.2f\n", salary);
    printf("Bonus = Rs. %.2f\n", bonus);
    printf("Tax = Rs. %.2f\n", tax);
    printf("Final Salary = Rs. %.2f\n", finalSalary);

    return 0;
}
```

```
C:\Users\HP\Documents\kji.e:
Enter basic salary: Rs. 50000
Enter years of service: 6

Basic Salary = Rs. 50000.00
Bonus = Rs. 7500.00
Tax = Rs. 0.00
Final Salary = Rs. 57500.00

--------------------------------
Process exited after 7.723 seconds with return value 0
Press any key to continue . . .
```

CamScanner

[*] kji.cpp

```cpp
#include <stdio.h>

float calculateLoan(float salary, float payments)
{
    float remaining, loan;

    remaining = salary - payments;
    loan = remaining * 10;

    return loan;
}

int main()
{
    int age;
    float salary, payments, remaining, loan;

    printf("Enter your age: ");
    scanf("%d", &age);

    printf("Enter monthly salary: Rs. ");
    scanf("%f", &salary);

    printf("Enter existing monthly loan payments: Rs. ");
    scanf("%f", &payments);

    remaining = salary - payments;

    if(age < 21)
        printf("Not eligible: Age must be at least 21.\n");

    else if(salary < 50000)
        printf("Not eligible: Salary must be at least Rs. 50000.\n");

    else if(payments < 0 || payments > salary)
        printf("Invalid loan payment amount.\n");

    else if(remaining < salary * 0.40)
        printf("Not eligible: Less than 40% of salary remains.\n");

    else
    {
        loan = calculateLoan(salary, payments);

        printf("\nYou are eligible for a loan.\n");
        printf("Remaining monthly income = Rs. %.2f\n", remaining);
        printf("Maximum loan amount = Rs. %.2f\n", loan);
    }
```

C:\Users\HP\Documents\kji.e:

```
Enter your age: 25
Enter monthly salary: Rs. 60000
Enter existing monthly loan payments: Rs. 30000

You are eligible for a loan.
Remaining monthly income = Rs. 30000.00
Maximum loan amount = Rs. 300000.00

--------------------------------
Process exited after 73.62 seconds with return value 0
Press any key to continue . . .
```

CamScanner

kji.cpp

```c
#include <stdio.h>
int isPalindrome(int n)
{
    int original, reverse = 0, digit;
    original = n;
    while(n > 0)
    {
        digit = n % 10;
        reverse = reverse * 10 + digit;
        n = n / 10;
    }

    if(original == reverse)
        return 1;
    else
        return 0;
}

int isPrime(int n)
{
    int i;

    if(n < 2)
        return 0;

    for(i = 2; i < n; i++)
    {
        if(n % i == 0)
            return 0;
    }

    return 1;
}

int main()
{
    int n;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    if(n % 2 == 0)
        printf("The number is Even.\n");
    else
        printf("The number is Odd.\n");

    if(isPalindrome(n))
        printf("The number is a Palindrome.\n");
    else
        printf("The number is not a Palindrome.\n");

    if(isPrime(n))
        printf("The number is Prime.\n");
    else
        printf("The number is not Prime.\n");
```

C:\Users\HP\Documents\kji.e

```
Enter a positive integer: 565
The number is Odd.
The number is a Palindrome.
The number is not Prime.

--------------------------------
Process exited after 7.33 seconds with return value 0
Press any key to continue . . .
```

CamScanner

kji.cpp

```cpp
#include <stdio.h>
float calculateCharge(float weight)
{
    if(weight <= 2)
        return 200;
    else if(weight <= 5)
        return 400;
    else
        return 700;
}

int main()
{
    float weight, distance;
    float base, distanceCharge = 0, discount = 0, total;
    int member;

    printf("Enter parcel weight in kg: ");
    scanf("%f", &weight);

    printf("Enter delivery distance in km: ");
    scanf("%f", &distance);

    printf("Are you a member? (1 = Yes, 0 = No): ");
    scanf("%d", &member);

    if(weight <= 0 || distance < 0 || (member != 0 && member != 1))
    {
        printf("Invalid input!\n");
        return 0;
    }

    base = calculateCharge(weight);

    if(distance > 10)
        distanceCharge = (distance - 10) * 20;

    total = base + distanceCharge;

    if(member == 1 && total > 1000)
        discount = total * 0.10;

    total = total - discount;

    printf("\nBase Charges = Rs. %.2f\n", base);
    printf("Distance Charges = Rs. %.2f\n", distanceCharge);
    printf("Discount = Rs. %.2f\n", discount);
    printf("Final Amount = Rs. %.2f\n", total);

    return 0;
```

C:\Users\HP\Documents\kji.e

```
Enter parcel weight in kg: 10
Enter delivery distance in km: 30
Are you a member? (1 = Yes, 0 = No): 1

Base Charges = Rs. 700.00
Distance Charges = Rs. 400.00
Discount = Rs. 110.00
Final Amount = Rs. 990.00

--------------------------------
Process exited after 10.95 seconds with return value 0
Press any key to continue . . .
```

CamScanner

[*] kji.cpp

```c
#include <stdio.h>
void showBalance(float balance)
{
    printf("Current Balance = Rs. %.2f\n", balance);
}
int main()
{
    float balance = 20000, amount;
    int choice;
    do
    {
        printf("\n--- ATM MENU ---\n");
        printf("1. Check Balance\n");
        printf("2. Deposit Money\n");
        printf("3. Withdraw Money\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                showBalance(balance);
                break;

            case 2:
                printf("Enter deposit amount: ");
                scanf("%f", &amount);

                if(amount > 0)
                {
                    balance = balance + amount;
                    printf("Deposit Successful!\n");
                    showBalance(balance);
                }
                else
                    printf("Invalid Amount!\n");
```

CamScanner

[*] kji.cpp

```
39
40              break;
41
42          case 3:
43              printf("Enter withdrawal amount: ");
44              scanf("%f", &amount);
45
46              if(amount <= 0)
47                  printf("Invalid Amount!\n");
48
49              else if(amount > 10000)
50                  printf("Maximum withdrawal is Rs. 10000.\n");
51
52              else if(amount > balance)
53                  printf("Insufficient Balance!\n");
54
55              else
56              {
57                  balance = balance - amount;
58                  printf("Withdrawal Successful!\n");
59                  showBalance(balance);
60
61                  if(balance < 5000)
62                      printf("Warning: Low Balance!\n");
63              }
64
65              break;
66          case 4:
67              printf("Thank you!\n");
68              break;
69
70          default:
71              printf("Invalid Choice!\n");
72      }
73
74  } while(choice != 4);
75
76  return 0;
77 }
```

CamScanner

```
ak;


ntf("Enter withdrawal amount: ");
nf("%f", &amount);

amount <= 0)
 printf("Invalid Amount!\n");

e if(amount > 10000)
 printf("Maximum withdrawal is Rs. 10000.\n");

e if(amount > balance)
 printf("Insufficient Balance!\n");

e

 balance = balance - amount;
 printf("Withdrawal Successful!\n");
 showBalance(balance);

 if(balance < 5000)
     printf("Warning: Low Balance!\n");


ak;

ntf("Thank you!\n");
ak;

:
ntf("Invalid Choice!\n");


!= 4);
```

```
C:\Users\HP\Documents\kji.e

2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 1
Current Balance = Rs. 20000.00

--- ATM MENU ---
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 2
Enter deposit amount: 5000
Deposit Successful!
Current Balance = Rs. 25000.00

--- ATM MENU ---
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 3
Enter withdrawal amount: 4000
Withdrawal Successful!
Current Balance = Rs. 21000.00

--- ATM MENU ---
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 4
Thank you!

--------------------------------
Process exited after 38.72 seconds with return value 0
Press any key to continue . . .
```

CamScanner

kji.cpp

```c
int main()
{
    float price, subtotal;
    float discount1 = 0, discount2 = 0;
    float fee = 0, total;
    int quantity, member, payment;

    printf("Enter product price: ");
    scanf("%f", &price);

    printf("Enter quantity: ");
    scanf("%d", &quantity);

    printf("Are you a member? (1=Yes, 0=No): ");
    scanf("%d", &member);

    printf("Payment method (1=Cash, 2=Card): ");
    scanf("%d", &payment);

    if(price < 0 || quantity <= 0 ||
       (member != 0 && member != 1) ||
       (payment != 1 && payment != 2))
    {
        printf("Invalid Input!\n");
        return 0;
    }
    subtotal = calculateSubtotal(price, quantity);

    if(subtotal > 5000)
        discount1 = subtotal * 0.10;

    if(member == 1)
        discount2 = subtotal * 0.05;

    total = subtotal - discount1 - discount2;

    if(payment == 2)
        fee = total * 0.02;

    total = total + fee;

    printf("\n--- SHOPPING BILL ---\n");
    printf("Subtotal = Rs. %.2f\n", subtotal);
    printf("10%% Discount = Rs. %.2f\n", discount1);
    printf("Member Discount = Rs. %.2f\n", discount2);
    printf("Payment Fee = Rs. %.2f\n", fee);
    printf("Final Amount = Rs. %.2f\n", total);

    return 0;
}
```

```
C:\Users\HP\Documents\kji.e:
Enter product price: 5500
Enter quantity: 2
Are you a member? (1=Yes, 0=No): 1
Payment method (1=Cash, 2=Card): 1

--- SHOPPING BILL ---
Subtotal = Rs. 11000.00
10% Discount = Rs. 1100.00
Member Discount = Rs. 550.00
Payment Fee = Rs. 0.00
Final Amount = Rs. 9350.00

--------------------------------
Process exited after 13.69 seconds with return value 0
Press any key to continue . . .
```

CamScanner